

Q4. (Tracing the Pseudocode) [5 Marks]

Trace the following pseudocode step by step for the input list:. Show the value of all variables at each iteration. What are the final values of total and odd_count?

```
BEGIN
SET total = 0
SET odd_count = 0
FOR i = 1 TO 6
INPUT num
IF num MOD 2 == 1 THEN
total = total + num
odd_count = odd_count + 1
ELSE
total = total - num
ENDIF
ENDFOR
OUTPUT total, odd_count
END
```

Step-by- Step Trace Table

For the input list: [8, 12, 5, 20, 3, 15]

Iteration (i)	Input (num)	Check (num MOD 2 == 1)	total Calculation	total Value	odd_count Value
Initial	-	-	-	0	0
1	8	False (Even)	0 - 8	-8	0
2	12	False (Even)	-8 - 12	-20	0
3	5	True (Odd)	-20 + 5	-15	1
4	20	False (Even)	-15 - 20	-35	1
5	3	True (Odd)	-35 + 3	-32	2
6	15	True (Odd)	-32 + 15	-17	3

Final Values

- **total:** -17
- **odd_count:** 3